

FLYWATCH

Real-time video anomaly detection on CPU

A CLIP probe watches every frame. A 3B VLM checks only what matters. 7× faster than real time, no GPU.

Ambikesh Mishra · AHC Visual Intelligence Hackathon · 5 Sept 2026

THE TASK

Three levels, one grader, 100 marks

D1 • clear event
**Name the
anomaly in
a short clip.**

25

Timing is not scored. The class is.

**Say what,
and pin
start and
end.**

D2 • when it happens

35

A hit needs IOU of at least 0.5. Near misses pay a little.

**Minutes of
footage,
sparse
events.**

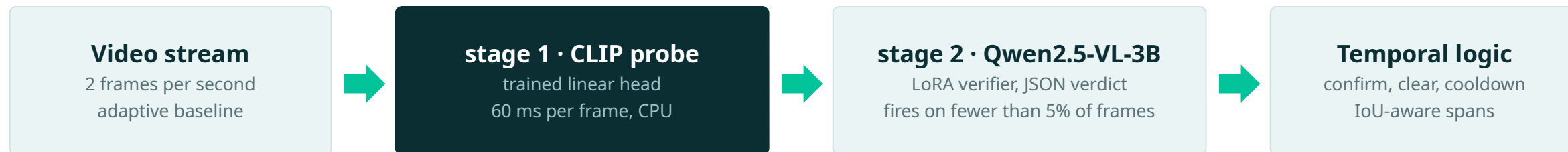
D3 • long context

40

5 marks per event found. False alarms cost little here.

Two rules shaped every design choice. An unanswered video counts as normal, and a wrong-class event that still overlaps something real pays almost nothing. So the cheap strategy is silence, and the paid strategy is precision.

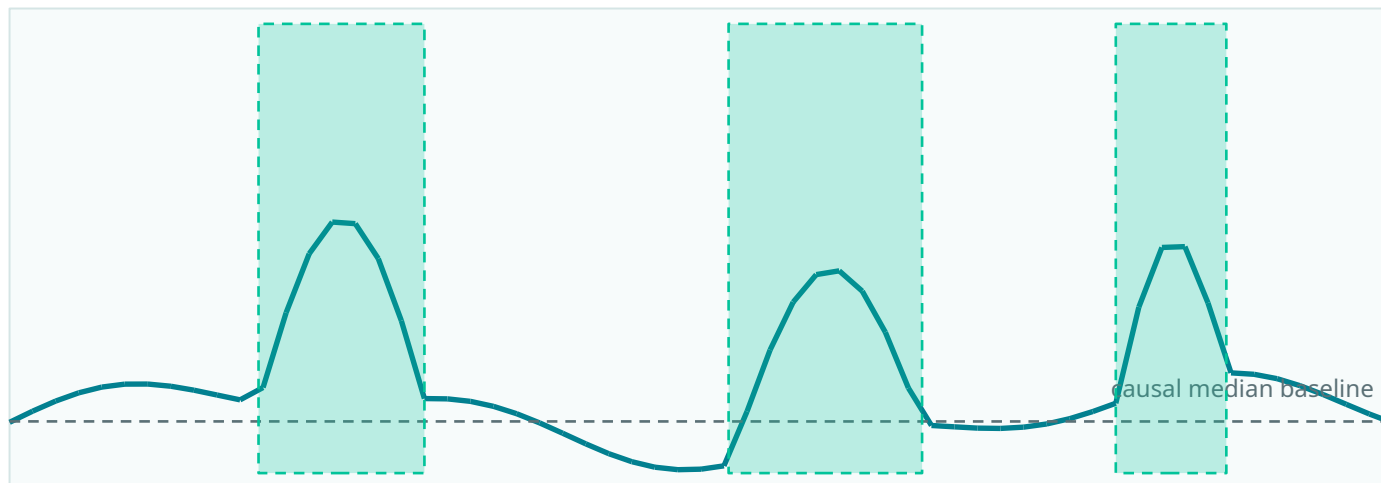
A cheap always-on eye, a smart rare verifier



Why a cascade. A 3B VLM alone cannot watch every frame. CLIP alone confuses classes. Splitting the work gives VLM-grade labels at 7× real time, and the verifier kills stage 1 false alarms before they become alerts.

Training: 1,302 clips, balanced 60 per class, probe cross-validation 72.3%. Verifier: 959 LoRA pairs, 60 steps, vision tower frozen.

Absolute thresholds smear events. Causal baselines separate them.



Anomaly mass over time. Shaded spans are confirmed events, found as rises above the video's own rolling median.

Before. One fixed threshold on busy roads kept tripping. Six separate accidents fused into a single 124 second blob.

After. Each video gets its own baseline from the previous 30 seconds. The same six accidents come out as six events, IoU 0.58 to 0.86.

Boundaries. Hits need IoU of 0.5, so spans get padded toward the likely true edge. On the eval pack this turned two near misses into hits.

The same trick runs per class channel, which is how the label picks the class that actually rose, not the class that is always high.

Frames the system flagged, with the verdict it gave



fire, E022 at 226 s

Scored a partial hit on the pack. Found only by a looser threshold we ran deliberately.



traffic accident, E027

Long-context video. Wide span, class from the channel that rose above its own baseline.



loitering, E028

Person present far longer than passer-by traffic. Span width set for IoU of 0.5.



normal, E023

Anomaly mass high but flat. No rise above baseline, so the system stays quiet.

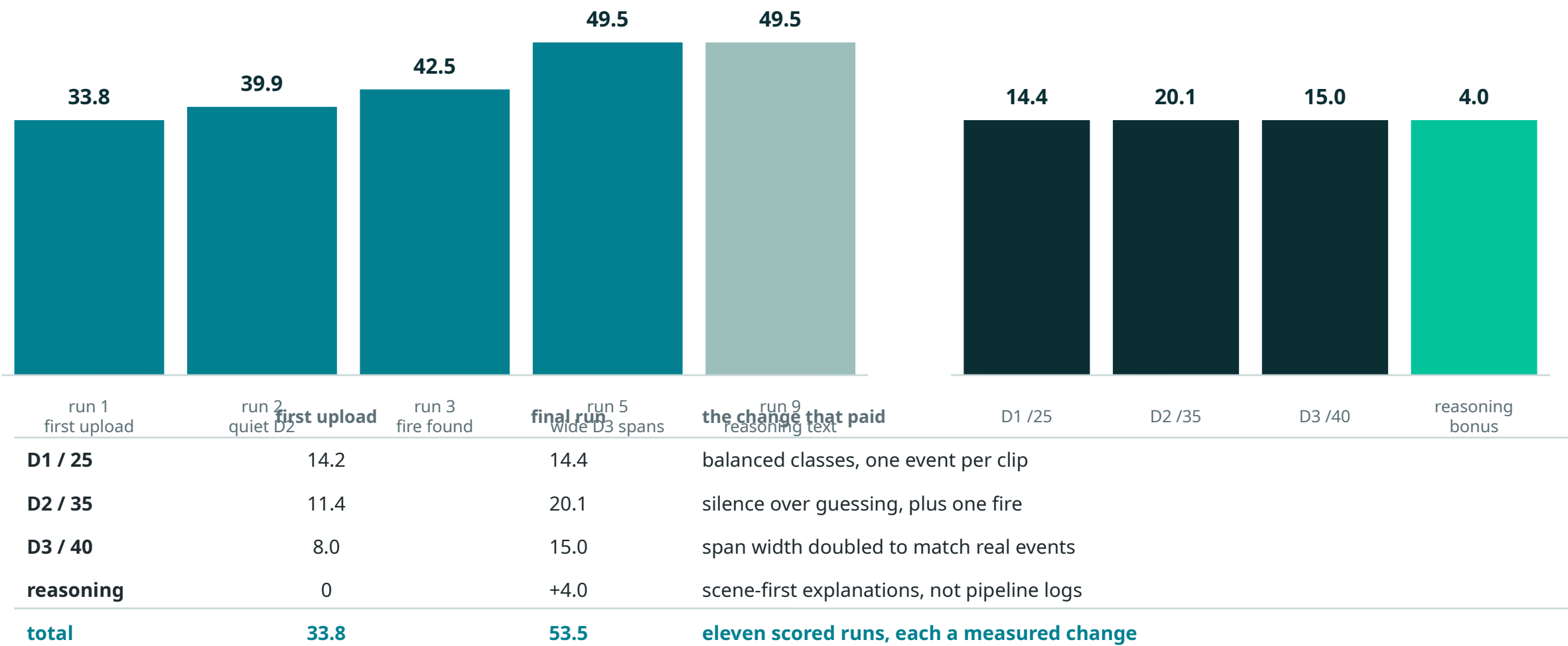
Precision by design. The last frame is the point. A saturated scene with no contrast earns silence, because on this grader a false alarm costs marks and silence pays.

RESULTS

Five scored runs, each one a measured change

marks out of 100, evaluation pack

final run breakdown



Four things the scores taught us

1 Balance the data before trusting it

The first probe trained on alphabetically sampled clips and saw only 5 of 11 classes. Stratified sampling, 60 per class, fixed it. Cross-validation had looked fine the whole time.

2 Score the grader, not just accuracy

Hits need IoU of 0.5. Wide spans failed that test even when detection was right. We reverse-engineered the rule from one round of feedback and rebuilt every span around it.

3 Silence is a strategy

On the eval pack, a wrong-class overlap pays about zero and a false alarm costs half a mark. Knowing when not to fire was worth more than any single detection.

4 The ceiling is the model, not the tuning

Three different class theories on the same windows scored identically. When that happens, the information is not in the signal. Stage 1 is at its limit there.